

DATABASE MANAGEMENT SYSTEMS (DBMS) COURSE PROJECT

IT Helpdesk & Asset Support Management System

Presentation-III: Live User Interface Demonstration & MySQL Relational CRUD Operations

PRESENTED BY

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Section: AIML Panthers | Serial No: 39

Presentation Slot: Wednesday, 7 October 2026 | 9:30 AM – 9:40 AM

ACADEMIC GUIDANCE

Dr. Kiranmayee Adavala

Faculty Guide | School of Technology

Woxsen University, Hyderabad

Live Architecture: MySQL Community 9.x + Flask + Vanilla SPA

Enterprise IT Support & Asset Tracking Dilemma

OPERATIONAL BOTTLENECKS

Fragmented Channels

- Enterprises manage hundreds of devices and support requests via chaotic emails and spreadsheets.
- Untracked tickets lead to missed SLAs and zero technician accountability.
- Hardware relocations occur without updating ownership or tracking warranty expiries.
- Lack of immutable audit logging leads to untracked status modifications.

RELATIONAL SOLUTION

Normalized Database

- A centralized MySQL 8.0+ relational back-end (it_helpdesk) enforcing 3NF normalization.
- Clear subtype separation into Incidents vs Service Requests.
- Complete audit trail of status progressions and technician assignments.
- Real-time responsive web UI connected to MySQL demonstrating live CRUD.

Objectives & Scope of Presentation–III

CORE OBJECTIVES

Core System Objectives

- Deliver a fully working User Interface connected directly to MySQL (no mock data).
- Demonstrate real-time INSERT of ticket records reflected in database tables.
- Demonstrate permanent DELETION of records with referential cascade.
- Demonstrate multi-table VIEW of records with dynamic filtering and audit timelines.

TECHNICAL SCOPE

Scope & Verified Deliverables

- 14 Relational tables managed with foreign key cascades and restrict constraints.
- Live system health monitoring with active connection verification.
- Interactive 3D database visualizer inspecting schema nodes and relationships.
- SQL verification playground executing live analytical queries for faculty.

Database Architecture & 3NF Relational Model

SCHEMA STRUCTURE

14 Normalized Tables

- Master Entities: departments, users, categories, priorities, support_staff, warranties.
- Operational Entities: assets, tickets, incidents, service_requests, assignments.
- Audit & Service Entities: status_histories, resolutions, maintenance.
- Storage Engine: MySQL InnoDB with ACID transaction support and foreign keys.

INTEGRITY CONSTRAINTS

Enforced Relational Rules

- ON DELETE RESTRICT: departments → users, categories → tickets, priorities → tickets.
- ON DELETE CASCADE: tickets → incidents, service_requests, assignments, status_histories, resolutions.
- ON DELETE SET NULL: users → assets, warranties → assets.
- CHECK constraints on priority levels (1–5), status enums, and positive maintenance costs.

Full-Stack System Architecture & Decoupled Design

BACK-END & DATABASE

MySQL & Python REST API

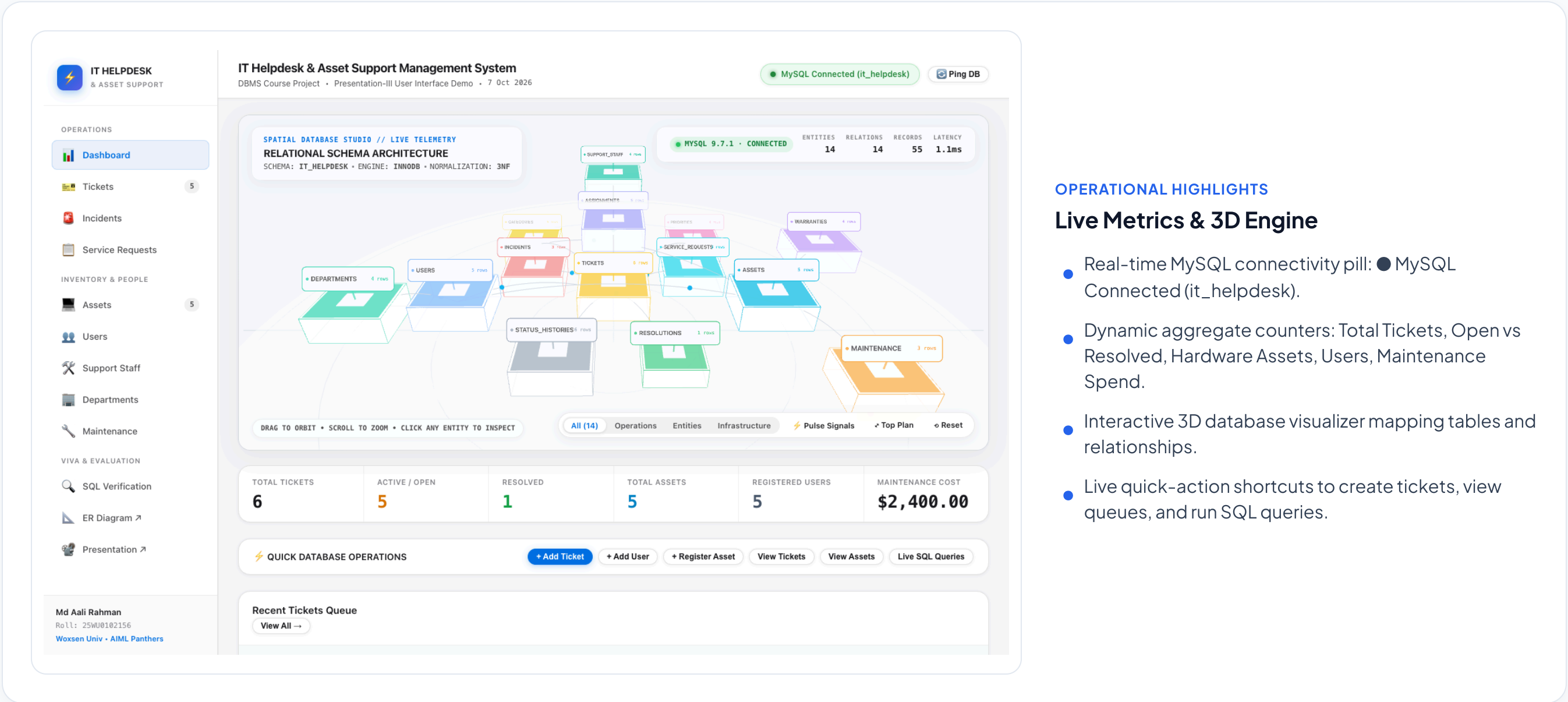
- Database Server: MySQL 8.0+ / 9.x running locally on port 3306.
- Driver: mysql-connector-python with parameterized queries (zero SQL injection).
- Backend: Python Flask REST API server exposing CRUD endpoints.
- Transaction Management: Explicit autocommit=False with atomic commits and rollbacks.

FRONT-END PRESENTATION

Modern Single-Page Application

- Vanilla HTML5 / CSS3 / ES6+ JavaScript client using the native Fetch API.
- Zero heavy frontend framework dependencies for high performance.
- Three.js 3D WebGL database relationship visualizer.
- Live SQL verification playground for faculty evaluation.

Operational Command Dashboard



VIEW: Multi-Table Relational Ticket Queue

IT HELPDASK
& ASSET SUPPORT

OPERATIONS

Dashboard

Tickets5

Incidents

Service Requests

INVENTORY & PEOPLE

Assets5

Users

Support Staff

Departments

Maintenance

VIVA & EVALUATION

SQL Verification

ER Diagram↗

Presentation↗

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MySQL Connected (it_helpdesk)Ping DB

Helpdesk Tickets Management

Query, filter, inspect audit logs, insert new records, and delete tickets with real-time MySQL persistence.

+ Add TicketRefresh

Search ticket #, title, user, category...

All StatusesAll CategoriesAll Types

TICKET #	TITLE	TYPE & SUBTYPE	REQUESTER	DEPARTMENT	PRIORITY	STATUS	CREATED AT	ACTIONS
TKT-006	yo Access & Permissions	Service Request yo	Aarav Sharma aarav.sharma@company.com	Engineering	High	Open	2026-10-06 20:28	ViewUpdate
TKT-005	Request second monitor for software engineering	Service Request Hardware Allocation Asset Procurement	Vikram Singh vikram.singh@company.com	Engineering	Medium	Open	2026-08-23 08:45	ViewUpdate
TKT-004	Install Figma desktop client	Incident Software Application Issue Software & OS	Ananya Iyer ananya.iyer@company.com	Marketing & Sales	Low	Open	2026-08-22 16:30	ViewUpdate
TKT-003	Finance accounting portal connection timed out	Incident Network Outage Network & VPN	Rohan Verma rohan.verma@company.com	Finance & Accounts	Critical	In Progress	2026-08-22 14:00	ViewUpdate
VPN access								

RELATIONAL JOINS

Queues & Details

- Executes multi-table SQL JOIN across tickets, users, departments, categories, and priorities.
- Dynamic filtering by Status, Category, and Subtype (Incident vs Service Request).
- Real-time search across ticket number, title, requester name, and category.
- Detail modal displays full relational graph: requester details, assigned technicians, and status history.

INSERT: Form-Driven Record Creation & Subtyping

Create New Helpdesk Ticket

Requester (User) *

Select User...

Priority Level *

Select Priority...

Issue Category *

Select Category...

Ticket Classification *

☒ Incident (Failure / Outage)

☐ Service Request (Provisioning)

Incident Type *

e.g. Hardware Malfunction, System Crash, Network Outage

Will be inserted into the incidents relational subtype table.

Ticket Summary / Title *

Brief summary of the issue (e.g. Laptop display flickering)

Detailed Description *

Provide full context, symptoms, error messages, and troubleshooting steps...

Cancel

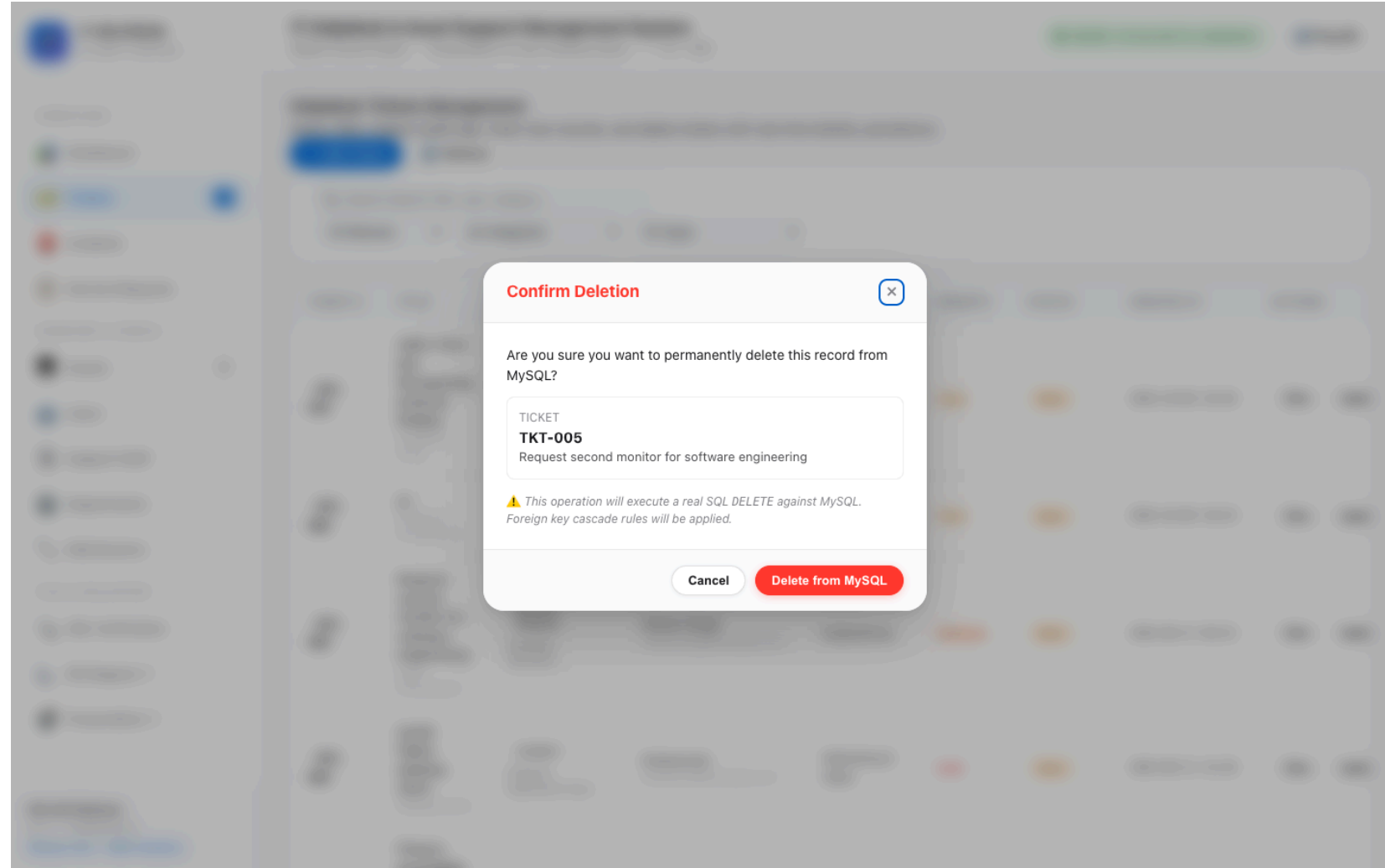
Create Ticket in MySQL

ATOMIC CREATION

Multi-Table Atomic Flow

- Dynamic foreign key dropdowns populated from live database tables.
- Automated sequential ticket tracking number generation (TKT-006).
- Atomic multi-table transaction inserting into tickets, subtypes, and status_histories.
- Immediate UI reflection: new row appears instantly in the table without page reload.

DELETE: Permanent Removal with Foreign Key Cascades



REFERENTIAL CASCADE

Permanent Record Purge

- Confirmation dialog displaying exact record ID prevents accidental deletions.
- Executes `DELETE FROM tickets WHERE ticket_id = %s` directly in MySQL.
- `ON DELETE CASCADE` automatically purges sub-entities without leaving orphan rows.
- `ON DELETE RESTRICT` safely protects users and categories with active dependencies.

Live Database Persistence & Refresh Verification

IT HELPDESK
& ASSET SUPPORT

OPERATIONS

Dashboard

Tickets 5

Incidents

Service Requests

INVENTORY & PEOPLE

Assets 5

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MySQL Connected (it_helpdesk)

Ping DB

SPATIAL DATABASE STUDIO // LIVE TELEMETRY

RELATIONAL SCHEMA ARCHITECTURE

SCHEMA: IT_HELPDESK • ENGINE: INNODB • NORMALIZATION: 3NF

MYSQL 9.7.1 • CONNECTED

ENTITIES	RELATIONS	RECORDS	LATENCY
14	14	55	1.1ms

ZERO MOCK DATA

ACID Storage Guarantee

- State persistence verified across browser refreshes (Cmd+R / F5).
- Direct terminal verification via MySQL client confirms record existence in InnoDB storage.
- Zero mock data, fake arrays, or localStorage used — 100% real database transactions.
- Demonstrates end-to-end ACID consistency.

Automated DBMS Test Suite Execution (TC01 – TC10)

CORE CRUD TEST CASES

CRUD & Persistence Tests

- TC01: Database Connection to MySQL — **PASS**
- TC02: View Tickets Relational Query — **PASS**
- TC03: Insert New Ticket Record — **PASS**
- TC04: Refresh & Persistence Verification — **PASS**
- TC05: Delete Ticket from MySQL — **PASS**
- TC06: Refresh After Delete (Remains Absent) — **PASS**

CONSTRAINT TESTS

Relational Constraints & Errors

- TC07: Form Input Validation (Missing Required Fields) — **PASS**
- TC08: Unique Constraint Enforcement (Duplicate Email) — **PASS**
- TC09: Foreign Key Deletion Violation (RESTRICT) — **PASS**
- TC10: MySQL Offline Error Handling (Safe User Message) — **PASS**
- **Overall Result: 10/10 Test Scenarios Passed (100% Success Rate)**

Conclusion & Future System Enhancements

SUMMARY OF OUTCOMES

Presentation–III Accomplishments

- Successfully delivered an enterprise-grade IT Helpdesk and Asset Support Management System.
- Live MySQL database connectivity verified with real-time INSERT, DELETE, and VIEW operations.
- Strict 3NF normalization and foreign key integrity constraints maintained throughout.
- All official presentation requirements fulfilled.

FUTURE ROADMAP

Future Enhancements

- Automated SLA escalation triggers and email notification webhooks.
- Multi-factor authentication (MFA) and Role-Based Access Control (RBAC).
- Predictive hardware maintenance forecasting using machine learning models on servicing history.